

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
4.	(a)	(i)		(annual saving) $12 \times 50 = £600$ (1) (Total insulation cost) $3\,800 + 2\,900 = £6\,700$ (1) Pay back time = $6\,700(\text{ecf})/600(\text{ecf}) = 11.17$ years (1) (so not true) OR (Total insulation cost) $3\,800 + 2\,900 = £6\,700$ (1) $6\,700(\text{ecf})/50 = 134$ months (1) $134(\text{ecf})/12 = 11.17$ years (1) (so not true) If ecf gives less than 10 years – so true (if the ecf calculation is correct only 2 marks) 1 mark for correct judgement based on incorrect answer. Accept: Total insulation cost = $£3\,800 + £2\,900 = £6\,700$ (1) Amount paid back in 10 years = $10 \times 12 \times 50 = £6\,000$ (1) £700 not paid off so not true / more needed to pay off cost (1)			3	3	3	
		(ii)		Any 3 × (1) from: Maintenance costs (1) Tilt/angle of panel/not facing the Sun (1) Cost of electricity may decrease (1) less sunshine hours than expected / more cloud (1) Panels are dirty (1) Electricity consumption decreases (1)		3		3		